INCH-POUND

MIL-DTL-21617/29
w/AMENDMENT 1
27 July 2017
SUPERSEDING
MIL-DTL-21617/29
24 July 2003

DETAIL SPECIFICATION SHEET

CONNECTOR, RECEPTACLE ELECTRICAL, RECTANGULER, POLARIZED SHELL, MINIATURE TYPE, BACK SHELLS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-21617.

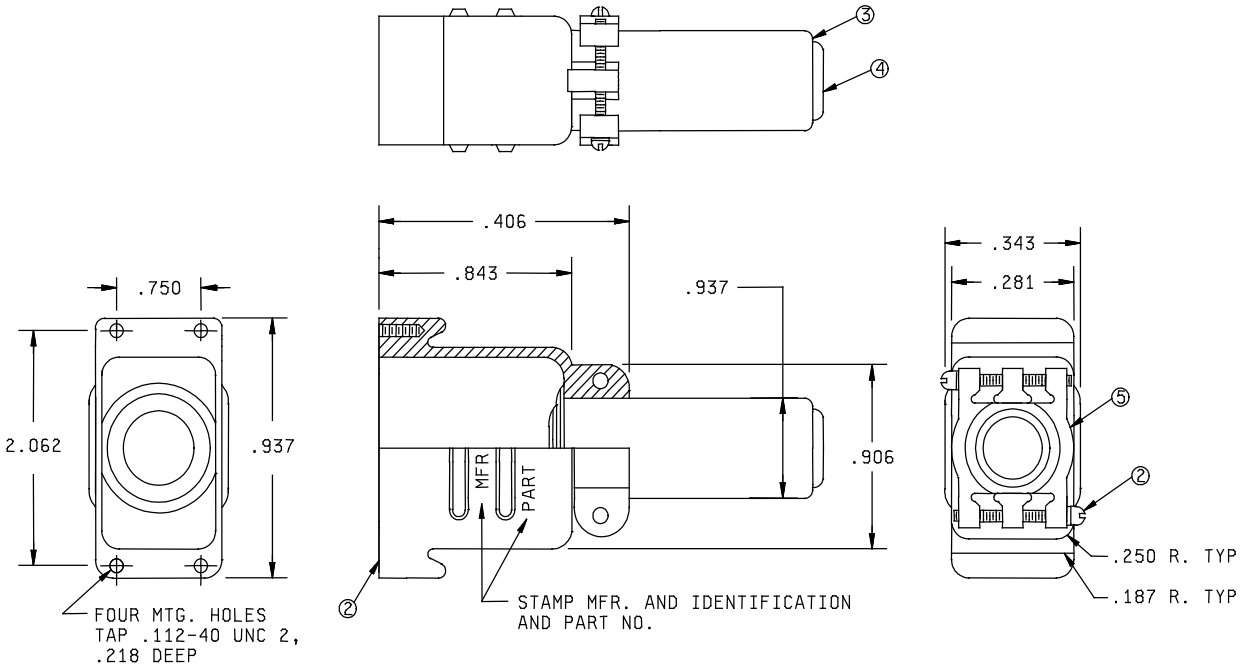


FIGURE 1. Junction shell, straight.

AMSC N/A

FSC 5935

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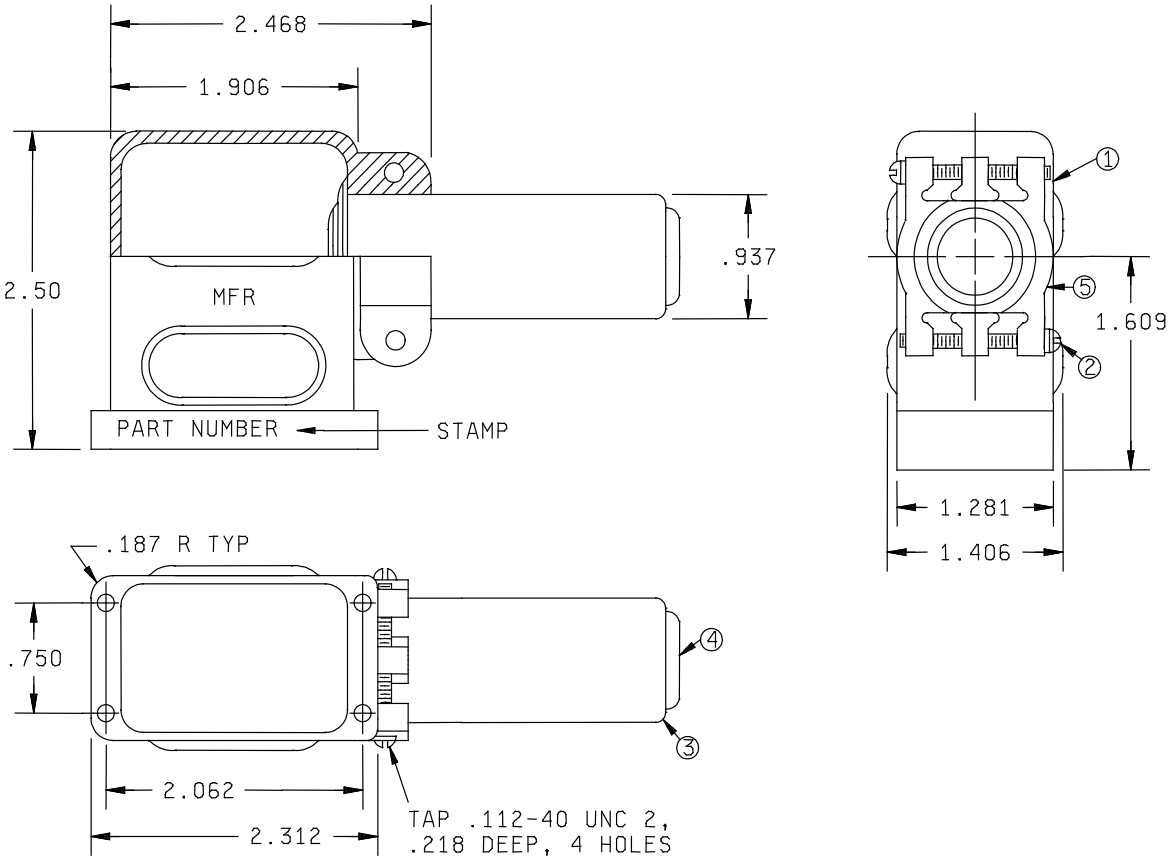
Item	Part	Number required
1	End bell 1.125 inches	1
2	.164-32 x 1.125 FIL HD machine screw	2
3	Bushing	1
4	Bushing	1
5	Clamp, gland	2

Inches	mm	Inches	mm	Inches	mm
.112	2.84	.281	7.14	.843	21.41
.187	4.75	.343	8.71	.906	23.01
.218	5.54	.406	10.31	.937	23.80
.250	6.35	.750	19.05	2.062	52.37

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).

FIGURE 1. Junction shell, straight - Continued.

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Item	Part	Number required
1	End bell 1.125 inches	1
2	.164-32 x 1.125 FIL HD machine screw	2
3	Bushing	1
4	Bushing	1
5	Clamp, gland	2

Inches	mm	Inches	mm
.112	2.84	1.281	32 54
.164	4.17	1.406	35.71
.187	4.75	1.609	40.87
.218	5.54	1.906	48.41
.750	19.05	2.062	52.37
.937	23.80	2.312	58.73
1.125	28.58	2.468	62.69
		2.50	63.50

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).

FIGURE 2. Junction shell, 90 degree angle.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2.

Reference material: See MIL-DTL-21617.

Shells:

Material: Shells shall be made from high-grade aluminum alloy conforming to SAE-AMS-QQ-A-250/8, temper H-32.

Finish: Cadmium plate in accordance with SAE-AMS-QQ-P-416, type II, class 3. The resultant finish shall be electrically conductive.

Screws:

Material corrosion resisting, type AISI 302, 303, 304, 305, 316, 384 (developed for cold heading) in accordance with SAE-AIR4127, or type UNS S30430 in accordance with the chemical requirements of ASTM A493.

Treatment: Passivation shall be in accordance with SAE AMS2700.

Magnetic permeability: These screws shall have a magnetic permeability or 2.0 maximum (air = 1.0) for a field strength of H = 200 oersteds, using a magnetic indicator in accordance with ASTM A342/A342M.

Threads: Threads shall be in accordance with FED-STD-H28.

Threads: Screws shall have complete threads extending to within two threads of the head, or closer if practicable.

Part or Identifying Number (PIN): PIN example as follows:

M21617/29	-	A
Specification number		Shell style

Shell Style: A - 90 degree
S - Straight

Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to MIL-DTL-21617, this document references the following:

FED-STD-H28
ASTM A342/A342M
ASTM-A-493
SAE-AMS-QQ-A-250/8
SAE-AMS-QQ-P-416
SAE-AIR4127
SAE AMS2700

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2017-060)

Review activities:
Army - AV, MI

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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